

ROSHAN KUMAR GUPTA

Mohali, Punjab, India

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Career Objective

Motivated Computer Science Engineering student with a strong foundation in low-level systems architecture, high-performance computing, and deterministic runtimes. Deeply passionate about building robust, physics-aware solutions and scalable platforms that solve complex engineering challenges. Seeking a Research Engineer or Systems Architect Intern role where I can leverage my proficiency in C++, Python, and parallel computing architectures to drive innovation and push the boundaries of modern computational frameworks.

Education

Bachelor of Engineering in Computer Science (2025 – 2029)
Chandigarh University, Mohali – CGPA: 6.98 (First Year)
Relevant Coursework: High-Performance Computing, Low-Level Systems, Parallel Architectures, Database Systems.
Senior Secondary Education: CBSE Board (2023 – 2025)
RSVM – Percentage: 65%
Secondary Education: Matriculation CBSE Board (2022 – 2023)
GGPS – Percentage: 80.4%

Work Experience

Earthos Platform | Author & Lead Architect (Jul 2026 – Present)
• Authored *The AGI Question: What Does It Take to Become a Mind?*; built storefront and reader.
• Engineered automated multi-warehouse nearest-node shipping algorithms via Shiprocket APIs.
• Developed a secure PDF generation pipeline with real-time DRM watermark fingerprinting.
SMG Electric Scooters Ltd | L3 IT Intern (Dec 2025 – Jan 2026)
• Designed an internal CRM portal, streamlining logistics and vendor workflows via automation.
Chandigarh University | Student Representative (Oct 2025 – Sep 2026)
• Managed high-stakes communication to resolve academic and infrastructure issues for 400+ students.

Projects

Project TARS Ecosystem | Tools: Python, Data Science (2025)
• Secured \$5,500 Emergent Ventures grant; designed the TARS Core physics-constrained validation framework achieving 70.1% precision on NASA TESS data.
• Engineered *tars_infra*, a parallel FITS data ingestion engine utilizing multi-drive spanning to index 150,000+ files from the MAST database.
FieldChain | Tools: C++, Vulkan (2025)
• Secured \$1,500 grant; engineered a GPU-centric storage layer sustaining 7GB/s sequential writes via PCIe 4.0, outperforming specialized DPUs by 4.2x via Vulkan.
Project Earthos (Synapse Arch) | Tools: Python, AI (2026)
• Architected a neural-symbolic cognitive substrate combining event-sourced memory hydration.
• Implemented AST locks to block non-deterministic floating-point namespaces, enforcing execution parity.

Technical Skills

Low-Level: CUDA, Vulkan, PTX, C++, Deterministic Runtimes, Memory Architectures
Systems & ML: React (Architecture), PyTorch (Physics-ML), Next.js, Figma (UI Primitives)
Backend & Tools: Node.js, PostgreSQL, Supabase, REST APIs, Git, GitHub, VS Code

Interpersonal Skills

- Leadership & Collaboration:** Proven capability in managing student cohorts (400+) and leading hackathon teams.
- Problem Solving:** Strong analytical approach to systems architecture and hardware-software integration.
- Technical Advocacy & Liaison:** Experienced in high-stakes communication between academic admins and student groups.

Achievements

- 1st Place, Tekathon 2026 (SIH Internal):** Smart Vehicles category for ISRO PS (ID: SIH26168).
- 4th Place, Hacktoberfest 2026:** Global open-source software contributions.
- Awarded Best Student Paper for FieldChain at IEEE CCNCPS 2026 conference in Dubai. (2026)
- Secured cumulative research grants totaling \$6,500 (Emergent Ventures, FieldChain). (2025)
- Winner, ZeroToOne Hackathon (DeFi Treasury Dashboard) & Zinnovatio Hackathon.

Certifications

- Research Excellence Award – Chandigarh University. (May 2026)
- Scientific Writing – Polytechnique. (Oct 2025)
- Azure AI Cert (Vision, NLP) – Microsoft. (Feb 2026)

Area of Interests and Hobbies

Technical Hobbies: Systems Architecture & High-Performance Computing, Quantum Noise Limits & NISQ Circuits, Astronomical Data & Physics-ML Integration, Hardware Tinkering & Competitive Programming.
Languages Known: English (Professional Working), Hindi (Native or Bilingual).